Project Synopsis

E-Learning Platform

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[2025 –26]

YASHWANTRAO BHONSALE INSTITUTE OF TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled "**E-learning platform**'' is a bona fide work of **Mr.Fendre Nikhil Prasad (CS15), Mr.Naik Harshad Arjun (CS36) , Mr.Parulekar Utkarsh Vishwanath. (CS39) , Mr.Satardekar Atharv Omprakash (CS49)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **"Bachelor of Engineering"** in **"Computer Science Engineering"**

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YASHWANTRAO BHONSALE INSTITUTE OF TECHNOLOGY

Project Report Approval

This Mini Project Report entitled "**E-learning platform** " by **Mr.Fendre Nikhil Prasad (CS15), Mr.Naik Harshad Arjun (CS36) , Mr.Parulekar Utkarsh Vishwanath.(CS39) ,Mr.Satardekar Atharv Omprakash (CS49).** is approved for the degree of **"Bachelor of Engineering in Computer Science Engineering"** 2025-2026.

Examiners

1……

(Internal Examiner Name & Sign)

2……

(External Examiner Name & Sign)

Place: YBIT, Sawantwadi

Date:

Declaration

We declare that this written submission represents my ideas in my own words and where others ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Fendre Nikhil Prasad (Cs15)	Naik Harshad Arjun (Cs36)	Parulekar Utkarsh Vishwanath (Cs39)	Satardekar Atharv Omprakash (Cs49)

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1. Introduction

The rapid advancement of internet technology has transformed the traditional education system into a digital learning environment. Online learning platforms have become popular because they provide flexibility, accessibility, and convenience to learners. However, most existing platforms allow automatic enrollment and unrestricted course access after registration. This reduces institutional control and makes it difficult to monitor student performance effectively.

The proposed project, E-Learning Management System with Admin-Controlled Course Access, is a web-based application developed using the MERN stack (MongoDB, Express.js, React.js, Node.js). The system is designed to provide a structured and monitored online learning environment. In this system, students can view available courses but can access them only after receiving approval from the administrator or teacher.

The platform includes features such as course management, enrollment approval, progress tracking in percentage format, online assessment, and certificate generation after qualification. This system ensures academic discipline, transparency, and performance-based certification. It is suitable for colleges, coaching centers, and training institutes where proper monitoring and structured learning are required.

2.Literature Survey

A detailed study of existing Learning Management Systems (LMS) and online education platforms was conducted to understand current technologies and limitations. Many platforms provide online course delivery and assessment features, but most lack proper enrollment control and structured monitoring mechanisms.

Existing research on Learning Management Systems highlights the importance of modular course structures, secure authentication systems, progress tracking mechanisms, and online assessment modules. Studies show that progress tracking improves course completion rates, and secure login systems protect user data from unauthorized access.

Research on cloud-based LMS platforms indicates improved scalability and accessibility. Additionally, studies on online examination systems emphasize transparency and automated evaluation methods to ensure fair certification processes.

Based on the literature survey, it is observed that while many systems provide digital learning features, there is a need for a simplified platform that integrates enrollment approval, structured monitoring, assessment validation, and certification into a single system.

Literature Survey Table:

Paper / Source	Key Finding	Relevance to My Project
Study on Modular LMS Design	Modular systems improve content management	Used to design course modules in React
Research on Secure Authentication Systems	Authentication enhances data privacy	Implemented JWT-based login security
Online Assessment System Research	Automated testing ensures fair evaluation	Used for final test module
Cloud-Based LMS Research	Cloud systems improve scalability	Future deployment planning
Progress Tracking in E-learning	Progress monitoring increases completion rates	Implemented percentage-based progress tracking

3.Problem Statement & Motivation

Online learning platforms often provide immediate course access without approval or structured monitoring. This leads to uncontrolled enrollments, limited supervision, and certificate issuance without strict evaluation. Educational institutions require a system where administrators can control access, monitor progress, and ensure that students qualify before receiving certificates.

The motivation behind this project is to develop a secure and structured online learning system that maintains academic integrity while providing flexibility.

3.1 Problem Statement

Current online learning systems lack administrative control over course enrollment and monitoring. Students gain direct access without verification, and certificates may be issued without strict qualification checks. There is a need for a system that ensures controlled access, tracks student progress accurately, and provides certification only after successful assessment.

3.2 Proposed Solution

The proposed solution is a MERN stack-based web application that introduces an enrollment request mechanism. Students must request enrollment in a course. The admin reviews the request and approves or rejects it. Only after approval can students access course materials.

The system automatically tracks student progress based on completed lectures. After completing all modules, students must attempt an online test. If they achieve the minimum passing score, the system generates a certificate.

The admin dashboard allows course management, enrollment approval, progress monitoring, and test management. This ensures transparency, control, and academic credibility.

4.Technology Stack Selection

The project uses the MERN stack, which provides full-stack JavaScript development and scalability.

4.1 Programming Language

JavaScript is used as the primary programming language. It enables both frontend and backend development, ensuring consistency and faster integration.

4.2 Framework

Node.js and Express.js are used for backend development. They support RESTful API creation, middleware integration, authentication handling, and scalable server-side processing.

4.3 Database

MongoDB is used as the database system. It stores user data, course details, enrollment records, progress information, test results, and certificates. MongoDB's flexible schema supports dynamic educational content.

4.4 Frontend

The frontend is developed using **React.js**. React provides a component-based architecture, making the application modular and scalable. It uses React Router for navigation and React Hooks for state management.

The frontend includes:

- Student Dashboard
- Admin Dashboard
- Course Listing Page
- Enrollment Request Page
- Progress Tracking Interface
- Test Module
- Certificate Page

Bootstrap and CSS are used for responsive and user-friendly interface design.

4.5 Development Tools

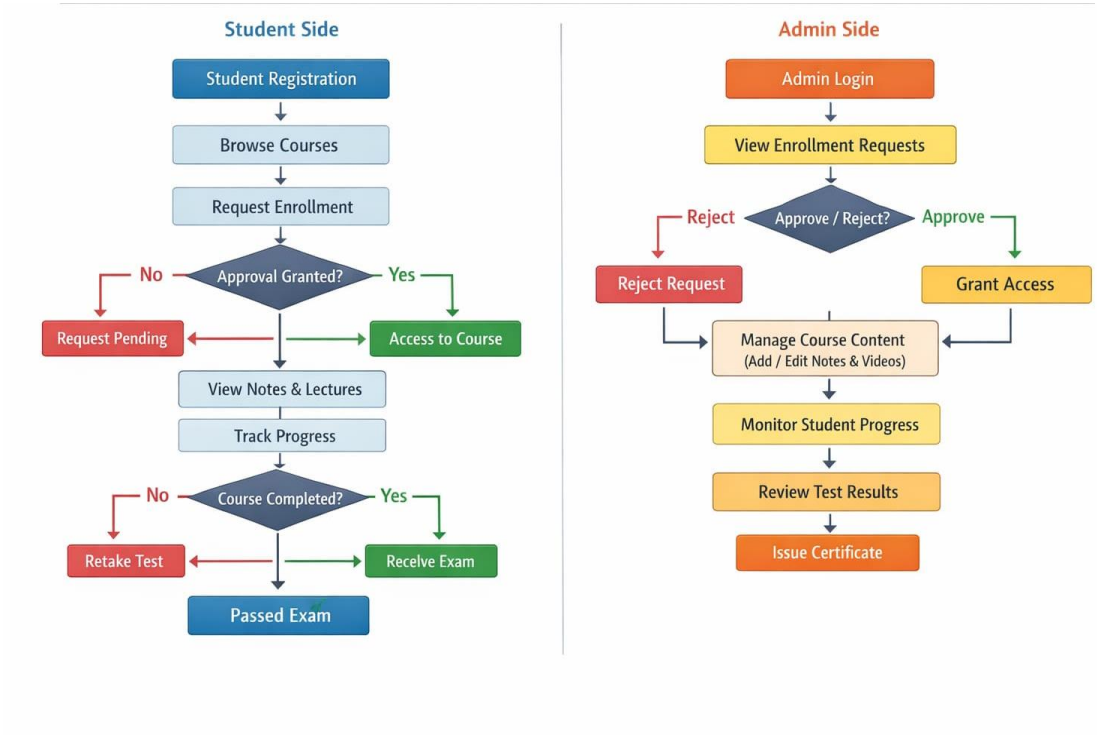
The tools used include:

- Visual Studio Code for development
- Postman for API testing
- MongoDB Compass for database management
- Git and GitHub for version control
- Chrome Developer Tools for debugging

5.System Modeling

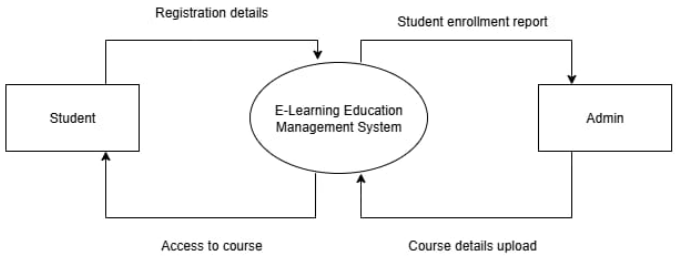
System modeling represents the structural and functional design of the E-learning Platforml++. It helps in understanding how the system works before actual implementation. This section includes diagrams that explain data flow and system operations.

5.1 Flowchart



5.2 Data Flow Diagram

Level 0 :



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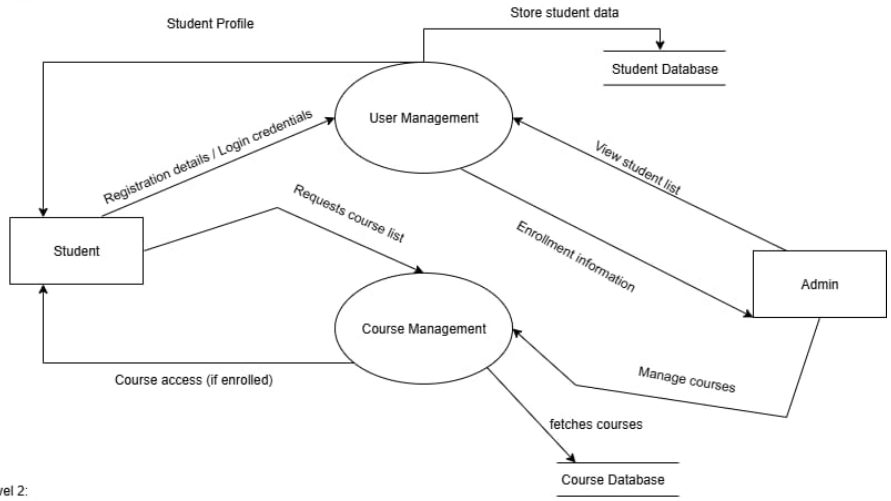

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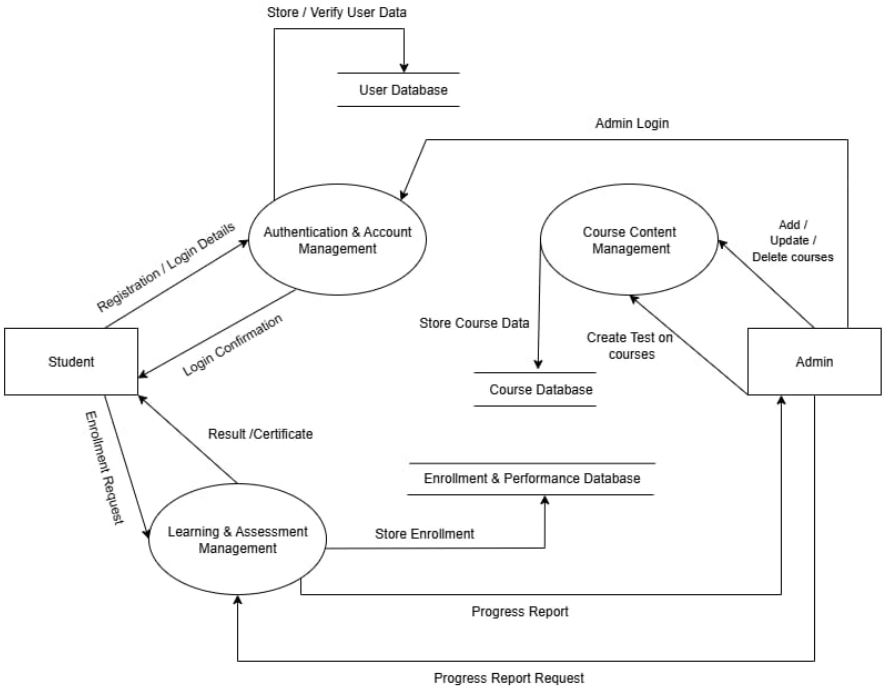

6.Conclusion and Future Work

The E-Learning Management System with Admin-Controlled Access provides a structured and disciplined online learning environment. It ensures controlled course access, accurate progress tracking, and qualification-based certification. The system enhances academic credibility and monitoring efficiency.

Future enhancements may include payment gateway integration, AI-based course recommendations, live video classes, discussion forums, mobile application development, cloud deployment, and advanced analytics dashboards.

References :

Research Papers / Journals

- J. Smith, “Online Learning Systems and Performance Evaluation,” International Journal of Computer Science, Vol. 12, Issue 4, pp. 45–52, 2021.

- Kumar, “Secure Authentication in Web Applications,” Journal of Information Security, Vol. 8, Issue 2, pp. 30–37, 2020.

Websites

- GeeksforGeeks, “Introduction to Learning Management Systems,” https://www.geeksforgeeks.org

- MongoDB Official Documentation, “MongoDB Basics,” https://www.mongodb.com

- Node.js Official Website, “Node.js Documentation,” https://nodejs.org